MIL-C-10859H
20 March 1984
SUPERSEDING
MIL-C-10859G
6 November 1973

MILITARY SPECIFICATION

CLOTH, OXFORD, COTTON, (PERMEABLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for permeable cotton cloth.

* 1.2 Classification. The cloth shall be of the following types and classes as specified (see 6.2 and 6.7):

Type I - 5.2 ounces

Class 1 - Dyed
Class 2 - Natural flame resistant treated
Class 3 - Bleached white

Type II - 6.0 ounces

Class 1 - Dyed
Class 3 - Bleached white

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon-Covered

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 Flame resistant finish approval (type I, class 2). The cloth shall be treated with an approved flame resistant treatment (see 6.5).

3.3 Material. The material shall be cotton which has been carded, drawn and spun into single yarn.

3.4 Color.

* 3.4.1 Types I and II, class 1. When dyed cloth is specified, the color of the types I and II, class 1 cloth shall be as specified (see 6.2), and shall match the standard sample (see 6.3). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

* 3.4.1.1 Types I and II, class 3 bleached white cloth. When bleached white cloth is specified, the cloth shall be fully bleached and may be supplemented with Fluorescent optical brighteners which Fluoresce to the blue region. The bleached cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.2.3. When no standard sample is available, the discoloration shall be no lower than the rating of "fair" when tested as specified in 4.2.3.

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* 3.4.2 Type I, class 2. The type I, class 2 finished cloth shall be the natural shade imparted to the undyed unbleached cloth, resulting from the treatment.

* 3.4.3 Matching. The color of the dyed cloth shall match the standard sample under artificial daylight having a correlated color temperature of 7000 ± 500 K and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K.

3.4.4 Colorfastness.

* 3.4.4.1 Types I and II, class 1. The types I and II, class 1 dyed cloth shall show fastness to laundering (after 3 cycles), perspiration, light and crocking equal to or better than the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed cloth shall show "good" fastness to laundering (after 3 cycles) perspiration and light, and shall show an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.2.3.

* 3.5 Physical requirements. The finished cloth shall conform to the applicable physical requirements specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Type and class	Weight ounces per sq. yd.	Yarns per inch (minimum)		Breaking strength (minimum) pounds		Tearing strength (minimum) pounds		Seam efficiency (minimum) percent
		Warp	Filling	Warp	Filling	Warp	Filling	
Type I:								
Class 1 and class 3	5.2 min	124	42	75	60	-	-	90
Class 2	6.8 max	122	40	75	60	3.0	3.0	80
Type II:								
Class 1 and Class 3	6.0 min	115	44	100	65	-	-	90

* 3.5.1 Width. The width of the cloth shall be as specified (see 6.2 and 6.4), and shall be the minimum acceptable width inclusive of selvages when fly shuttle looms are used, and exclusive of selvages and fringes when shuttleless looms are used.

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3.5.2 Weave. The weave shall be oxford, plain weave (two ends weaving as one).

3.6 Cloth preparation.

3.6.1 Types I and II, classes 1 and 3. The types I and II, classes 1 and 3 cloth shall be closely singed, desized and mercerized. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.6.2 Type I, class 2. The type I, class 2 cloth shall be desized. The starch and protein content including chloroform-soluble and water-soluble material of the cloth prior to flame resistant treatment shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.7 Finishing, type I, class 2 cloth. The cloth shall be treated with an approved durable flame resistant treatment (see 6.5). The finished cloth shall be thoroughly plant-washed to remove any residues from the curing process. The water-soluble material of the finished cloth shall not exceed 4.5 percent when tested as specified in 4.2.3.

3.7.1 Flame resistance. The finished cloth shall have an average time of after-flame of not more than 2.0 seconds and an average length of char of not more than 5.0 inches initially and after laundering (3 cycles), when tested as specified in 4.2.3.

3.7.2 Flexibility. The finished cloth shall have a flex stiffness of not more than 0.0008 inch-pound for both warp and filling when tested as specified in 4.2.3. In addition, the finished cloth shall have a stiffness of not more than 0.0035 inch-pounds for both warp and filling when tested as specified in 4.2.3 at $-25^{\circ}F \pm 2^{\circ}F$ after being subjected to that temperature for a period of 1 hour.

3.7.3 Air permeability. The air permeability of the finished cloth shall be no less than 20 cubic feet per minute per square foot when tested as specified in 4.2.3.

3.8 pH. The pH value of water extract of finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

3.9 Shrinkage. Types I and II, classes and 1 and 3 cloth shall be preshrunk and shall not shrink more than 1.0 percent in either the direction of the warp or filling when tested as specified in 4.2.3. The preshrinking process used shall not be identified by name or trademark either on the cloth or on the ticket or package. Type I, class 2 cloth shall not be preshrunk.

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* 3.10 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up on full width rolls as specified in 5.1.

3.11 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

* 3.12 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall conform to the quality of product established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of FED-STD-4, which are clearly

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noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.6), "G" for slubs and "E" for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 60 points. If one roll exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects, when present shall be scored four points for each yard in which they occur:

Width less than specified.
Baggy, ridgy, or wavy cloth.
Objectionable odor.
Poor dye penetration, mottled, streaky, or cloudy.
Treating compound uneven.
Not closely singed (classes 1 and 3 only).

4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

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4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

* 4.2.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. If any roll in the sample is off shade, shaded side to side, side to center, or end to end, or if any roll does not have the same appearance as the standard sample, the lot represented by the sample shall be rejected.

* 4.2.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Face identification missing from either or both ends.
Face identification on wrong side.
Preshrinkage process identified by name or trademark on cloth or ticket.
Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit for classes 1 and 3 cloth shall be 4 continuous yards full width of the finished cloth. The sample unit for class 2 cloth shall be 1/2 yard full width of the cloth prior to flame resistant treatment to determine weight and starch and protein content, including chloroform-soluble and water-soluble material; and 4 continuous yards full width of the finished cloth. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Presence of labile sulfur	3.4.1	2020
Discoloration (bleached white cloth)	3.4.1.1	5660 1/ 2/
Colorfastness to:		
Laundering (after 3 cycles)	3.4.4	5610 3/ 4/
Perspiration	3.4.4	5680
Light	3.4.4	5660
Crocking	3.4.4	5651
Weight	3.5	5041
Yarns per inch	3.5	5050
Breaking strength	3.5	5100
Tearing strength	3.5	ASTM D 1424
Seam efficiency	3.5	5110 5/
Weave	3.5.2	Visual 1/
Mercerization (types I and II, classes 1 and 3)	3.6.1	Microscopic examination for swelling of cotton fiber 1/
Starch and protein content including chloroform-soluble and water soluble material	3.6.1 and 3.6.2	2611
Water soluble material (type I, class 2)	3.7	2611 6/
Flame resistance (type I, class 2):		
Initial	3.7.1	5903
After three launderings	3.7.1	5556 and 5903
Flexibility (type I, class 2):		
Initial	3.7.2	5206
At $-25^{o} \pm 2^{o}F$	3.7.2	5206
Air permeability	3.7.3	5450
pH	3.8	2811
Shrinkage in laundering	3.9	5550

1/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

2/ Method 5660 shall be used, except that exposure of the specimen and standard shall be for 20 standard hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be

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rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration shall be considered satisfactory if it is not worse than the rating of "fair" indicated in Method 5660.

3/ The specimens must be dried after each of the 3 laundering cycles.

4/ On the color transfer cloth evaluation, only the stain on the cotton fibers of the color tranfer cloth shall be evaluated.

5/ The needle shall measure 0.040 ± 0.001 inch across the blade at the eye. The thread for all classes shall be polyester/cotton or rayon covered in accordance with MIL-T-43548, ticket No. 50, 2 or 3 ply for the needle and ticket No. 70, 2 or 3 ply for the looper.

6/ The testing shall be performed in accordance with the "water-soluble material" paragraph of Method 2611, the chloroform extract procedure shall be omitted.

* 4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

* 5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B or Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The types I and II, classes 1 and 3 cloths are intended for use in clothing and mittens. Type I, class 2 cloth is intended for use in tent liners.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and class required (see 1.2).
c. Color required (see 3.4.1).
d. Width required (see 3.5.1).
e. Length of roll if other than specified (see 3.10).
f. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Preferred width for type I, class 2 cloth. The preferred width of type I, class 2 cloth is 45 ± 1/2 inches.

* 6.5 Approved flame resistant treatment. Approval of the flame resistant treatment is the responsibility of U.S. Army Natick Research and Development Center, Natick, MA 01760, and is based on extensive tests including those for toxicity which are not set forth in this document. Because of the time required to evaluate and approve new treatments (approximately 6 months) only those chemical treatments already approved will be considered acceptable for the related procurement. Information pertaining to the approval of new treatments should be obtained from the U.S. Army Natick Research and Development Center. The list of approved treatments may be obtained from the contracting activity.

6.6 Fabric defect scales. Fabric Defect Replica Kits are available from:

Sears Roebuck and Company
Department 817, (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.7 Cross reference. Types I and II, class 1 cloth now classified as "Dyed" were previously classified as "Plain finish".

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6.8 Changes from previous issues. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8305-0465

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activity:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-10859H	Cloth, Oxford, Cotton (Permeable)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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